[INCH-POUND]

A-A-59296
July 31, 1998
SUPERSEDING
MIL-I-22266C
June 24, 1971

COMMERCIAL ITEM DESCRIPTION

INSULATING COMPOUND, ELECTRICAL (FOR FIELD SPLICING APPLICATIONS)

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE. This commercial item description covers two component, room-temperature curing, insulating compounds for field-splicing applications.

2. CLASSIFICATION. The insulating compound shall be of the following types as specified (see 6.2 and 6.4):

Type I	Epoxy resin
Class A	Packaged in film bags
Class B	Packaged in metal containers
Type III	Polyurethane resin
Class A	Packaged in film bags
Class B	Packaged in metal containers

3. SALIENT CHARACTERISTICS.

3.1 Materials. The insulating compound shall be a two-component system consisting of a thermosetting resin base of the room temperature curing epoxy type or elastomer of the polyurethane type and a curing agent. The component materials of the resin base and curing agent shall be such that the cured insulating compounds shall not be corrosive to the metals from which the electrical cables are made, nor shall the cured compound cause deterioration of jacketing materials used in cable construction.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Commander, Naval Air Warfare Center Aircraft Division, Code 414100B120-3, Highway 547, Lakehurst, NJ 08733-5100.

FSC 5970

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3.1.1 Unit of issue -types I and III, class A. The unit of issue shall consist of a bag made of flexible transparent high strength film. The bag shall have two compartments with an internally rupturable barrier separating the two compartments and shall be rupturable by manipulation of the bag without the aid of tools. The inner bag shall permit mixing of the bag with the hands after rupture of the internal barrier. The physical properties of the bag shall conform to the values shown in table I. Unless otherwise specified in the contract or order (see 6.2), the insulating compound shall be furnished in 2-3/4, 6-1/4, or 12-1/2 fluid ounce sizes. This unit of issue bag shall be further protected by placing the bag in a sealed transparent high strength bag.
The unit of issue inner bag shall meet the requirements of table I when tested in accordance with 4.7 and 4.8.

TABLE I. Test values for unit of issue inner bag.

Tests (Types I and III, Class A)	Test value and tolerance
Package strength (4.7): shall not leak	Pounds (min):
2-3/4 fl. oz. Size	200
6-1/4 fl. oz. Size	250
12-1/2 fl. oz. Size	300
Air inclusion (4.8):	Sq. inch (max):
2-3/4 fl. oz. Size	0.25
6-1/4 fl. oz. Size	0.35
12-1/2 fl. oz. Size	0.50

3.1.2 Unit of issue - types I and III, class B. Unless otherwise specified in the contract or order (see 6.2), the unit of issue for the insulating compound shall be a kit consisting of separate metal cans of one quart resin and one quart curing agent.

3.2 Primer. The use of a primer specified by the manufacturer to improve the adhesion of the compound to the substrate is permitted. Primer, when specified by the manufacturer, shall not be corrosive to the electrical metal cable or cause deterioration to the jacketing materials and shall be used in all tests where required for the insulating compound to conform to specification requirements.

3.3 Application. The insulating compound shall be used on electrical cables, for insulating ends against moisture, and electrically insulating cable splices. It shall be formulated to flow between conductors causing them to be sealed and insulated to preserve the electrical and physical properties of the cable against moisture, dirt, fumes, fungus, or other foreign materials.

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3.4 Physical properties. The physical properties of the insulating compound shall be as specified in table II.

3.5 Electrical properties. The electrical properties of the insulating compound shall be as specified in table III.

3.6 Shelf storage life. Types I and III material furnished in either class A or class B packages shall be capable of being stored for 1 year from the date of manufacture and meet the requirements of tables II and III, provided that the daily mean temperature of the ambient air at the storage location falls within the range of -40 °C to 55 °C (-40 °F to 131 °F).

Table II. Physical properties.

Test	Requirement		Test paragraph
	Type I	Type III	
Specific gravity (cured)	1.1-1.2	1.1-1.2	1/ ASTM-D792
Viscosity (cps) (mixed) (max.)	3,600	1,500	4.4
Gel time (minutes (100 gram samples)	8.0 – 13.5	14.5-20.0	4.5
Tensile strength (psi) (min)	2,500	400	ASTM-D651 Type III- ASTM-D412
Elongation (percent, min.)	2	170	ASTM-D651 Type III- ASTM-D412
Water absorption (percent, max.)	1	1	ASTM-D570
Shrinkage (percent, max.)	1.6	1	4.6
Hardness	70-90 (Shore D)	(60-70) (Shore A)	2/ ASTM-D2240

Notes to table II:

1/ Three specimens shall be used and the average of the three determinations shall be reported.

2/ The hardness of the test specimens shall be determined after the specimens have been allowed to cure for a maximum of seven days at 73 ± 2°F (23 ± 1°C) and a relative humidity of 50 ± 4 percent. The reading shall be taken after the durometer presser foot is in contact with the specimen for 15 seconds.

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TABLE III. Electrical Properties.

Test	Requirement		Test method
	Type I	Type III	
Dielectric strength (volts/mil 1/8-inch sample)	300 (min.)	300 (min.)	ASTM-D149
Dielectric constant (max.) 100 Hertz			1/ ASTM-D150
30 °C	4.0	6.0	
60 °C	5.0	7.5	
70 °C	6.5	--	
Dissipation factor (max.) 100 Hertz			1/ ASTM-D150
30 °C	0.03	0.25	
60 °C	0.08	0.15	
70 °C	0.30	0.05	
Volume resistivity, (min.) (23.1 ±1.1 °C) (ohm-cm)	3.3×10^{13}	4.0×10^{13}	2/ ASTM-D257

Notes to table III:

1/ One specimen shall be cast as specified in 4.3.2. Three disks or squares shall then be cut from the slab for measurement. Metal foil electrodes without guard rings shall be applied to the specimen by using a very thin film of petrolatum. The electrodes so applied shall be thoroughly rolled to exclude air. The average of three measurements at each temperature shall be reported.

2/ Four disk specimens, 3 inches in diameter and 1/8 inch thick, shall be prepared as specified in 4.3.1.1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

4.2 Market acceptability. The item offered must have been sold to the government or commercial market for at least one year.

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4.3 Test specimens.

4.3.1 Molds.

4.3.1.1 Mold for sheet specimens. The mold shall consist of two highly polished plates, 3/8-inch thick, coated with a release agent and spaced by 1/8-inch thick U-shaped spacers of polytetrafluoroethylene or any material to which the compound will not adhere. The mold assembly shall be clamped together with sufficient pressure to prevent leakage of the compound. A mold to produce 6-inch by 6-inch specimens is suggested. The resulting cast slabs shall then be cut into the test specimens required.

4.3.1.2 Mold for shrinkage tests. The mold for linear shrinkage shall be a mold which will cast a test specimen 4 inches long by 3/4 inch wide by 1-1/4 inches high. The tolerance for the dimensions of the length and width shall be 0.010 inch. There shall be no variations of the length and width of the cast specimen because of such causes as mold release, flashing not being removed, or other inaccuracies of the mold which would give false readings for these dimensions. A mold release shall be used to ensure that the test specimen will not adhere to the mold.

4.3.2 Casting. The compound ingredients shall be mixed in accordance with the manufacturer's directions, then poured into the molds. When the ingredients are mixed at temperatures above 23 °C, the molds shall be preheated approximately 10 °C above the mixing temperature before the mixture is poured into the molds. In pouring, precautions shall be taken to avoid the entrapping of air.

4.3.3 Curing of test specimens. Unless otherwise specified in the individual test method, specimens shall be cured for 48 ± 2 hours at 73 ± 2 °F (23 ± 1 °C) at a relative humidity of 50 ± 4 percent. All specimens shall be removed from the molds in 24 ± 2 hours, and the total cure time shall include the time the specimen is in the mold. Unless otherwise specified, cured specimens shall be tested within two hours from the end of the cure period.

4.4 Viscosity. The viscosity of the mixed resin and hardener shall be determined by means of a Brookfield viscosimeter or equal, according to the directions supplied by the manufacturer of the instrument, at a temperature of 25 ± 0.5 °C. The viscosimeter spindle and speed shall be according to directions supplied by the manufacturer of the instrument consistent with the viscosity of the material tested at a temperature of 25 ± 0.5 °C. The use of a 25 °C temperature allows a 2 °C rise over the 23 °C standard test temperature. This rise usually occurs as a result of stirring, mixing, and handling.

4.5 Gel time. Samples for the gel time shall be mixed in accordance with the manufacturer's instructions, except that immediately before mixing, the materials shall be stabilized at an ambient temperature of 25 ± 0.5 °C. The sample, consisting of a 100-gram mass, shall be poured into a 3-ounce paper cup immediately after mixing. The samples shall be placed in a

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clear plastic box (approximately 9 inches by 9 inches by 9 inches) for reduction of drafts. The gel time shall be reported as the time required for the insulating compound to become stringy, as determined by the raising and lowering of a 1/16 inch diameter applicator stick. The resin need not become completely solid for this gel point.

4.6 Linear shrinkage. Three specimens 4 inches long by 3/4 inch wide by 1-1/4 inches high shall be cast in a mold as specified in 4.3.1.2 and 4.3.2.

4.6.1 Dimensions. The length and width of each specimen shall be determined by averaging three determinations of each dimension. The percent linear shrinkage shall be calculated as follows:

$$\frac{(\text{Mold dimension} - \text{average specimen dimension}) \times 100}{\text{Mold dimension}}$$

This shall be done for both dimensions on each specimen. The percent of linear shrinkage shall be reported as the average of the six determinations.

4.7 Package strength test (class A only). The filled inner bag enclosed in its outer bag (see 3.1.1) shall be placed between two flat steel plates, 6 inches by 12 inches by 0.25 inch thick. This assembly shall be placed in a hydraulic press and loaded for not less than 1 minute to the load prescribed in table I for the package size under test. Following this test, the inner bag shall be removed form its outer bag and examined for rupture. Any leakage from the inner bag or transfer of contents form one compartment to the other shall be cause for rejection of the lot represented.

4.8 Air inclusion test (class A only). Following the test of 4.7, the inner bag shall be laid on a flat horizontal surface and allowed to rest until the included air bubbles, if any, float to the top surface of the bag. The combined area of the bubbles, as viewed through the top surface of the bag, shall not exceed the prescribed values of table I for the applicable size package.

5. PACKAGING.

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Copies of the American Society for Testing and Materials (ASTM) documents may be obtained from ASTM, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

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6.2 Ordering Data. Procurement documents should specify the following:

a. Title, number, and date of this commercial item description.
b. Type and class of insulating compound.
c. Unit of issue, if different from that specified in 3.1.1
d. Levels of packaging and packing.

6.3 National Stock Numbers (NSNs). The following is a list of NSNs assigned which correspond to this CID. The list may not be indicative of all possible NSNs associated with the CID.

NSN	Type
5970-00-752-5430	Type I, Class A

6.4 Part identification number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

AA 59296	-	X
CID Number		Type designator 1A = Type I, Class A 1B = Type I, Class B 3A = Type III, Class A 3B = Type IV, Class B

6.5 Cross reference. Type II material previously furnished under MIL-S-22266B (ASG) dated July 26, 1961 has been deleted. Type III, polyurethane material, was added to MIL-I-22266C.

MILITARY INTERESTS:

Custodians:
Army - CR
Air Force - 11

Preparing activity:
Navy - AS
(Project 5970-1169)

Review Activities:
Army - CR4, MI
Navy - SH
Air Force - 85, 99
DLA-GS

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